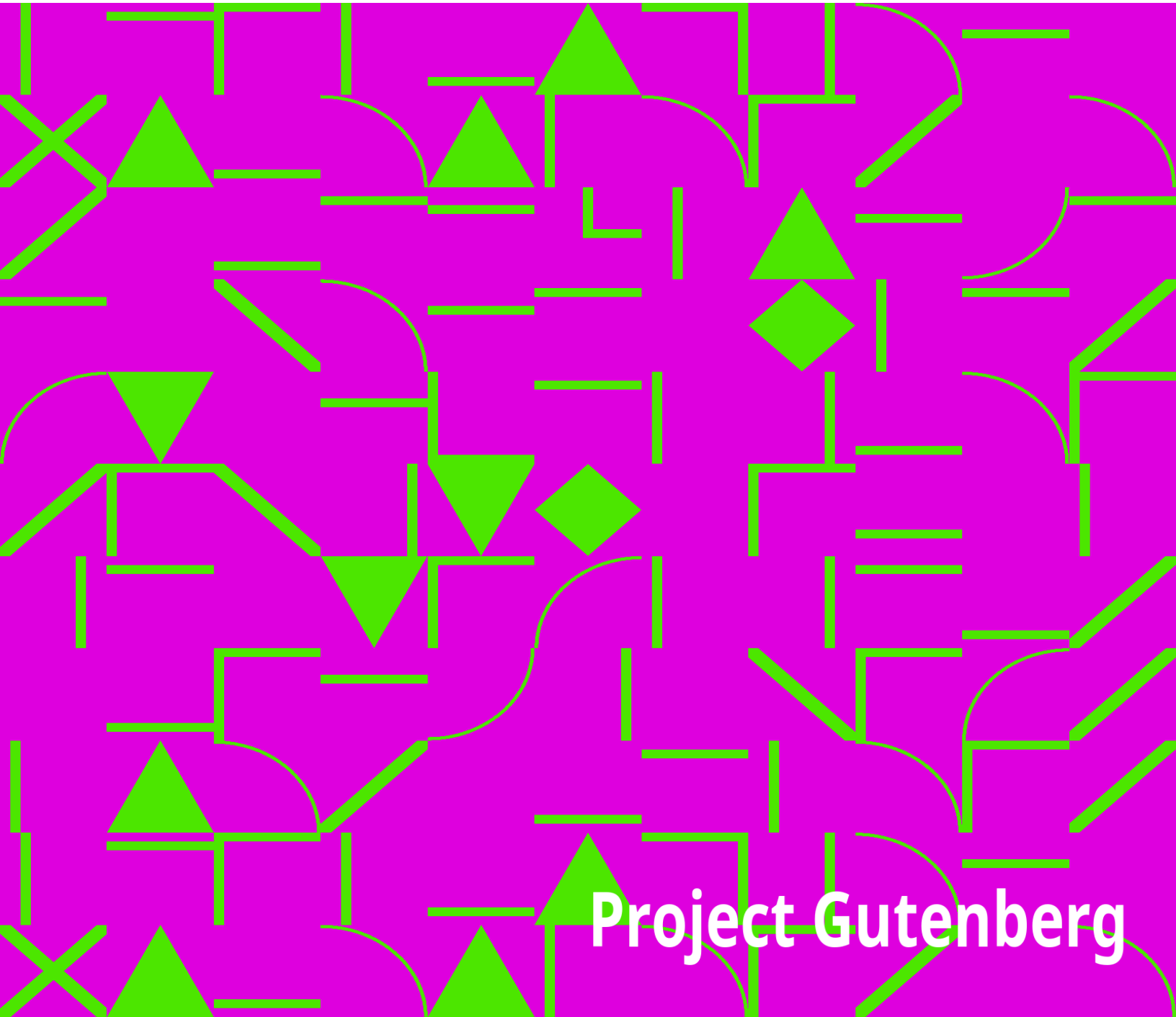


Geographic Variation in Red-backed Mice (Genus *Clethrionomys*) of the Southern Rocky Mountain Region

E. Lendell Cockrum and Kenneth Leonard



Project Gutenberg

Geographic Variation in Red-backed Mice (Genus *Clethrionomys*) of the Southern Rocky Mountain Region

E. Lendell Cockrum and Kenneth Leonard



Project Gutenberg

The Project Gutenberg eBook of Geographic Variation in Red-backed Mice (Genus Clethrionomys) of the Southern Rocky Mountain Region

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Geographic Variation in Red-backed Mice (Genus Clethrionomys) of the Southern Rocky Mountain Region

Author: E. Lendell Cockrum
Kenneth Leonard Fitch

Release date: June 4, 2010 [eBook #32679]

Language: English

Other information and formats: www.gutenberg.org/ebooks/32679

Credits: Produced by Chris Curnow, Tom Cos, Joseph Cooper and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK GEOGRAPHIC
VARIATION IN RED-BACKED MICE (GENUS CLETHRIONOMYS) OF
THE SOUTHERN ROCKY MOUNTAIN REGION ***

This file was derived from scanned images. The original text and copies of the included illustration is presented. Clicking on the Map will allow the user to view a larger copy of the scanned image from which it was derived.

Geographic Variation in Red-backed Mice (Genus Clethrionomys) of the Southern Rocky Mountain Region

BY

E. LENDELL COCKRUM and KENNETH L. FITCH

University of Kansas Publications
Museum of Natural History

Volume 5, No. 22, pp. 281-292, 1 figure in text

November 15, 1952

**UNIVERSITY OF KANSAS
LAWRENCE**

1952

**UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL
HISTORY**

**Editors: E. Raymond Hall, Chairman, A. Byron Leonard,
Edward H. Taylor, Robert W. Wilson**

**Volume 5, No. 22, pp. 281-292, 1 figure in text
November 15, 1952**

UNIVERSITY OF KANSAS
Lawrence, Kansas

PRINTED BY
FERD VOILAND, JR., STATE PRINTER
TOPEKA, KANSAS
1952



24-4369

Geographic Variation in Red-backed Mice (Genus *Clethrionomys*) of the Southern Rocky Mountain Region

BY

E. LENDELL COCKRUM and KENNETH L. FITCH

In the course of the preparation of a synopsis of the North American terrestrial microtines by one of us (Cockrum), and the completion of a Master's thesis on the geographical variation of the red-backed mice of Wyoming by the other (Fitch) we had occasion to study the red-backed mice of the southern Rocky Mountain region (see figure 1). Results of these studies are the recognition of two heretofore unnamed subspecies of the red-backed mouse in the southern Rocky Mountain region, and a clarification of the taxonomic status of two additional kinds.

***Clethrionomys gapperi galei* (Merriam)**

1890. *Evotomys galei* Merriam, N. Amer. Fauna, 4:23, October 8.

1931. *Clethrionomys gapperi galei*, Hall, Univ. California Publ. Zool., 37:6, April 10.

1897. *Evotomys gapperi galei*, Bailey, Proc. Biol. Soc. Washington, 11:126, May 13.

Type locality.--Ward, 9500 feet, Boulder County, Colorado.

Range.--The Rocky Mountains of extreme southern Alberta, Montana, northwestern and southern Wyoming, and north and central Colorado.

Remarks.--*C. g. galei*, with the largest geographic range of any of the Rocky Mountain subspecies, is also the most variable. Three principal areas of geographic variation were found. These areas are: The mountains of north-central Colorado and southern Wyoming (this area includes the type

locality); the Big Horn area probably northwest into Montana (no adult specimens from Montana or Alberta examined); and the Teton area which includes the mountains east and southeast of Yellowstone National Park. Specimens from these areas have noticeable differences in pelage, but no constant cranial differentiation could be detected. Specimens from the Medicine Bow Mountains of southern Wyoming have a more reddish dorsal stripe, and more buff and less gray on the sides than either of the northern geographic variants. The dorsal stripe continues farther anteriorly and is better defined through its entire length. There are fewer differences between the two northern geographic variants than between either one of them and the southern variant. Specimens from the Teton Mountains, however, have grayer sides, and the outer margin of the ear is tipped with chestnut (little or no chestnut shows on the ears of the specimens from the Big Horn Mountains); the dorsal stripe is less distinct (with slightly more gray throughout) than in either of the other geographic variants of the one subspecies.

Three specimens (two adults) are available from the Little Medicine Range in Converse County (22 miles south and 24.5 miles west of Douglas, 7600 feet), Wyoming. Although red-backed mice probably are found in the mountains of Natrona and Albany counties, the population, in the Little Medicine Range is somewhat isolated. In coloration these mice are lighter than any of the three geographic variants described above; the dorsal stripe is narrower; the sides are more buffy; the dorsal stripe does not project anteriorly beyond the ears as it does in the specimens from the Medicine Bow Mountains; and the face is grayer. These specimens resemble the population in the Big Horn Mountains to the north more than the population in the Medicine Bow Mountains.

The specimens from the Little Medicine Range, the Big Horn Range, and the Tetons are possibly subspecifically distinct from the southern specimens. Examination of specimens now allocated to *galei* from Montana and Alberta should aid in revealing whether the northern animals are an unnamed subspecies.

Specimens examined.--Total, 167, distributed as follows and unless otherwise stated, in the collection of the University of Kansas Museum of Natural History:

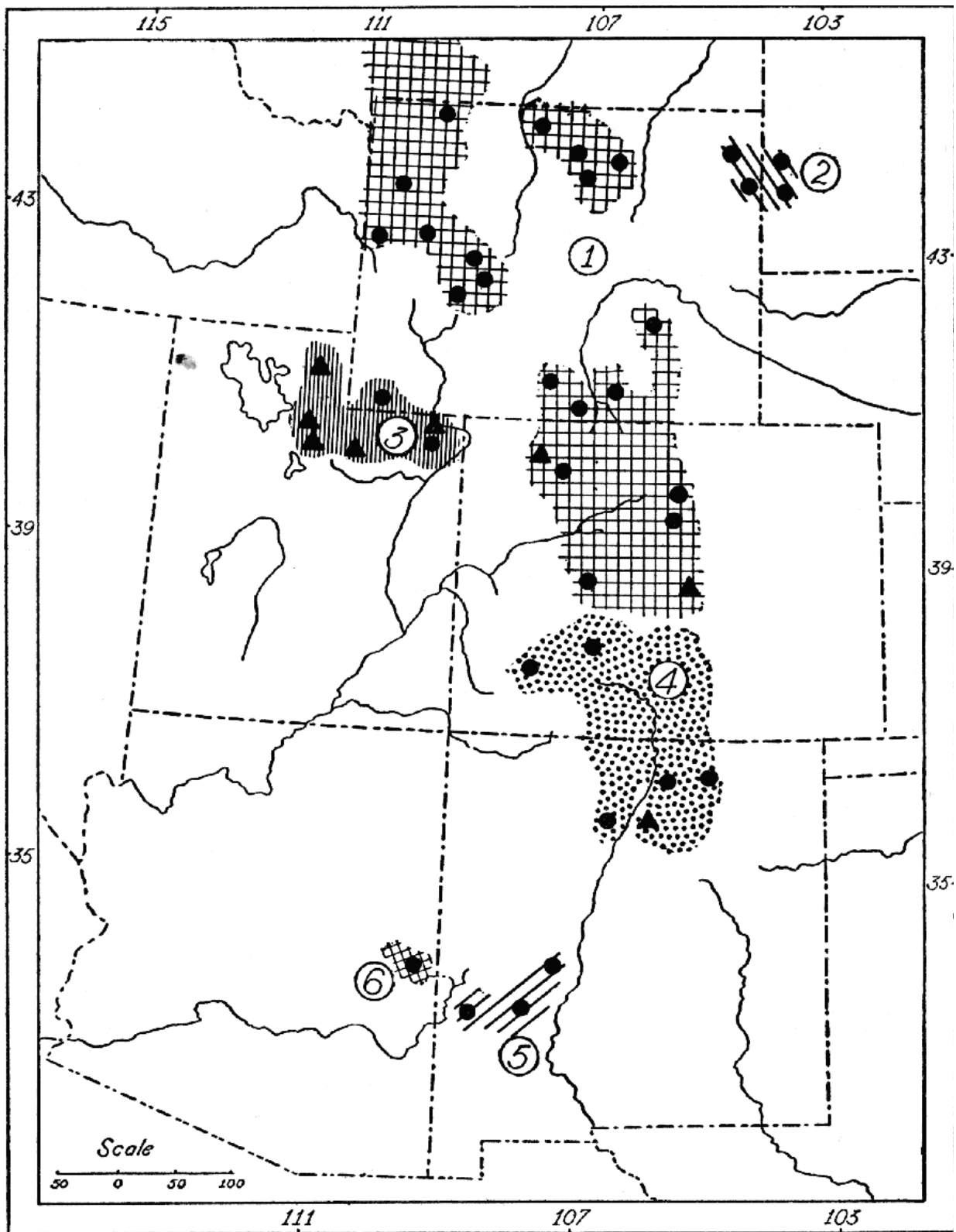
Wyoming: *Park County:* 28 mi. N and 3 mi. W Cody, 7200 ft., 1. *Big Horn County:* Medicine Wheel Ranch, 28 mi. E Lovell, 9000 ft., 22; 17-1/2 mi. E and 4-1/2 mi. S Shell, 1. *Teton County:*

Moran, 6244 ft., 4; Moran, 3 (James Findley Collection); 2-3/4 mi. E Moran, 6300 ft., 1; 3-3/4 mi. E and 1 mi. S Moran, 6200 ft., 10. *Washakie County*: 9 mi. E and 9 mi. N Tensleep, 8200 ft., 3; 9 mi. E and 4 mi. N Tensleep, 7000 ft., 1. *Johnson County*: 4 mi. W and 1 mi. S Klondike, 6500 ft., 1; 6-1/2 mi. W and 2 mi. S Buffalo, 5620 ft., 1. *Lincoln County*: 3 mi. N and 11 mi. E Alpine, 5650 ft., 1. *Sublette County*: 31 mi. N Pinedale, 8025 ft., 1. *Fremont County*: Togwotee Pass, 5 (James Findley Collection); 20-1/2 mi. W and 2 mi. S Lander, 1; Moccasin [=Moccasin] Lake, 19 mi. W and 4 mi. N Lander, 10,100 ft., 3; 18 mi. W and 3 mi. N Lander, 1; Mosquito Park Ranger Station, 17-1/2 mi. W and 2-1/2 mi. N Lander, 9500 ft., 10; 6-1/2 mi. W and 17 mi. S Lander, 8450 ft., 4; 5-1/2 mi. W and 22 mi. S Lander, 8800 ft., 3. *Converse County*: 22 mi. S and 24-1/2 mi. W Douglas, 7600 ft., 3. *Carbon County*: 18 mi. SW Rawlins, 7500 ft., 2; 19 mi. E and 8 mi. N Encampment, 9150 ft., 4; 19-1/2 mi. E and 6 mi. N Savery, 8800 ft., 1; 11 mi. E and 6 mi. N Savery, 8400 ft., 1; 14 mi. E and 6 mi. N Savery, 1. *Albany County*: 3 mi. ESE Browns Peak, 10,000 ft., 59.

Colorado: *Rio Blanco County*: 9-1/2 mi. SW Pagoda Peak, 7700 ft., 2. *Boulder County*: 2-1/2 mi. S Estes Park, 8400 ft., 2; 3 mi. S Ward, 8. *Clear*

Creek County: 2 mi. S Idaho Springs, 8000 ft., 1. *Gunnison County*: Gothic, 8 mi. N Crested Butte, 6 (James Findley Collection).

Additional records.--**Colorado:** *Rio Blanco Co.*: 25 mi. NE Meeker (Cary, N. Amer. Fauna, 33:120, 1911). *El Paso Co.*: Lake Moraine, 10,250 ft. (Warren, Mammals of Colorado, p. 224, 1942).



Click on map to see larger sized image.

FIG. 1.--Geographic ranges of the subspecies of *Clethrionomys gapperi* in the southern Rocky Mountains.

- | | | |
|-----------------------------|----------------------------|-----------------------------|
| 1. <i>C. g. galei</i> | 3. <i>C. g. uintaensis</i> | 5. <i>C. g. limitis</i> |
| 2. <i>C. g. brevipaudus</i> | 4. <i>C. g. gauti</i> | 6. <i>C. g. arizonensis</i> |
-

***Clethrionomys gapperi brevipaudus* (Merriam)**

1891. *Evotomys gapperi brevipaudus* Merriam, N. Amer. Fauna, 5:119, July 30.
1897. *Evotomys brevipaudus*, Bailey, Proc. Biol. Soc. Washington, 11:129, May 13.
1942. *Clethrionomys gapperi brevipaudus*, Bole and Moulthrop, Sci. Publ. Cleveland Mus. Nat. Hist., 5:153, September 11.

Type locality.--Three miles N Custer, 6000 ft., South Dakota.

Range.--The Black Hills of South Dakota and Wyoming.

Remarks.--Merriam (N. Amer. Fauna, 5:119, July 30, 1891) named this subspecies on the basis of two specimens collected in the Black Hills of South Dakota in July, 1888, and assigned it to the species *Evotomys* [= *Clethrionomys*] *gapperi*. He reported the diagnostic characteristics as: "Similar to *E. gapperi*, but with larger ears and shorter tail. The hazel of the dorsal area is not so bright as in *gapperi*; the sides are the same golden brown." Of the cranial and dental characteristics he wrote: "Much as in *E. gapperi*."

Bailey (Proc. Biol. Soc. Washington, 11:129, May 13, 1897), in his "Revision of the American voles of the genus *Evotomys*," with one additional specimen available, raised the Black Hills population to specific status, re-emphasizing the shortness of the tail, and pointing out a few slight cranial differences ("zygomatic arches low and flaring out, so that the inner instead of the outer side shows in top view; auditory bullae as large as in *gapperi*, but less rounded").

Bailey (*loc. cit.*) remarked that: "though based on so scanty material, the characters distinguishing the species are fairly pronounced. Its range is isolated and widely separated from that of any other members of the genus by open prairie country and a wide belt of the Transition zone. There seems to be no valid reason for considering it a subspecies."

Additional specimens have been taken in recent years from the Black Hills of South Dakota and northeastern Wyoming. This material has shed light on the relationships and morphological characteristics of the red-backed mice of this region. Bole and Moulthrop (*Sci. Publ. Cleveland Mus. Nat. Hist.*, 5:153, September 11, 1942) listed, as comparative material, eight specimens from Bull Springs, Custer County, South Dakota, under the name *Clethrionomys gapperi brevicaudus* (Merriam). They gave no reason for arranging *brevicaudus* as a subspecies of *C. gapperi*.

Twenty adults (11 skins and skulls, 9 skulls only) from Pennington County, South Dakota (specimens in the University of Michigan Museum of Zoology), have the following measurements (averages of external measurements based on 11 specimens only): Total length, 142 (123-155); tail, 35 (30-39); hind foot, 19.5 (18.6-21.0); basal length, 23.3 (21.7-24.5); condylobasilar length, 23.3 (21.9-24.5); zygomatic breadth, 13.7 (12.9-14.7); lambdoidal breadth, 11.7 (11.3-12.9); alveolar length upper cheek-teeth, 5.5 (5.2-5.8); interorbital breadth, 3.9 (3.6-4.1); length of nasals, 7.7 (7.1-8.5); breadth of rostrum, 3.2 (2.9-3.6); and length of incisive foramina, 5.0 (4.6-5.3).

Measurements of the type and one "more fully adult topotype" (as given by Bailey, *op. cit.*) are: Total length, 125, 130; tail length, 31, 32; hind foot, 19, 19; basal length, 21.2, 21.8; length of nasals, 6.6, 7.0; zygomatic breadth, 12.5, 12.8; mastoid breadth, 11.3, 11.0; alveolar length of upper molar series, 5.4, 5.3. In every measurement the figures for Bailey's specimens are smaller than the average of the same measurement in the 20 adults from Pennington County, and, in most measurements, are even lower than the minimum of the latter series. Therefore, we conclude that the material available to Merriam (*op. cit.*) and Bailey (*op. cit.*) consisted of only subadults.

In comparison with a series of 23 adult *Clethrionomys gapperi galei* from 28 mi. E Lovell, Big Horn County, Wyoming, *C. g. brevicaudus* has a

slightly shorter tail, longer hind foot, greater basal and condylobasilar lengths, greater zygomatic and lambdoidal breadths and conspicuously longer nasals.

In comparison with three adult *C. g. loringi* from Elk River, Sherburne County, Minnesota, *C. g. brevicaudus* has a greater total length, longer hind foot, greater basal length, conspicuously greater zygomatic and lambdoidal breadths, much longer nasals, and a narrower rostrum.

Clethrionomys gapperi brevicaudus, although isolated geographically and although morphologically more distinct than many of the currently recognized subspecies of *C. gapperi*, is probably best arranged as a subspecies of *C. gapperi* rather than as a full species. In certain characters, such as interorbital breadth and breadth of rostrum, it is intermediate between *C. g. galei* and *C. g. loringi*, but it resembles *C. g. galei* more than it does any other named kind.

Specimens examined.--Total, 66. Unless otherwise indicated, specimens from Wyoming are in the University of Kansas Museum of Natural History and specimens from South Dakota are in the University of Michigan Museum of Zoology. Specimens are distributed as follows:

Wyoming: *Crook County:* 3 mi. NW Sundance, 5900 ft., 3. *Weston County:* 1-1/2 mi. E Buckhorn, 6150 ft., 21; 12 mi. SE Newcastle, 1 (Univ. Michigan).

South Dakota: *Pennington County:* 1/2 mi. E Rochford, 1; 17 mi. NW Custer, 1; 16 mi. NW Custer, 20; 16 mi. SW Rapid City, 1; 3 mi. SE Hill City, 2; 4 mi. SE Hill City, 13; 5 mi. SE Hill City, Harney Peak, 7240 ft., 1. *Custer County:* 1-1/2 mi. E Sylvan Lake, 1.

***Clethrionomys gapperi uintaensis* Doult**

1897. *Evotomys gapperi galei*, Bailey, Proc. Biol. Soc. Washington, 11:127 (part from Uinta Mts. of Wyoming), May 13.

1941. *Clethrionomys gapperi uintaensis* Doult, Proc. Biol. Soc. Washington, 54:161, Dec. 8.

Type locality.--Paradise Park, 10,050 feet, 45 miles by road northwest Vernal, Uintah County, Utah.

Range.--The Uinta Mountains of northern Utah and southwestern Wyoming.

Remarks.--From the description given by Dutt (Proc. Biol. Soc. Washington, 54:161, 1941) in his original description of this subspecies, it appears that he had available for comparisons, only subadult specimens of *C. g. galei*. As judged from the material of *C. g. uintaensis* available to us (1 topotype, KU 38081, and 7 specimens from Uinta County, Wyoming, listed below) and from Dutt's (*op. cit.*) description and measurements, the subspecies *C. g. uintaensis* is but weakly differentiated from *C. g. galei*. No marked cranial differences are evident between the two subspecies; the differences in pelage noted by Dutt (*op. cit.*:161), however ("Similar to *Clethrionomys gapperi galei* from Ward, Colorado, but head and cheeks grayer; sides and back paler; belly whiter."), do seem to be valid.

On the basis of these differences in pelage and the geographic isolation of the range, we judge that *uintaensis* should be retained as a subspecies of *C. gapperi*. It is clear, however, that *C. g. uintaensis* is less distinct from *C. g. galei* than are the other adjacent subspecies.

Specimens examined.--Total, 8, all in the University of Kansas Museum of Natural History, distributed as follows:

Wyoming: *Uinta County:* 9 mi. S Robertson, 8000-8400 ft., 3; 9 mi. S and 2 mi. E Robertson, 8000 ft., 2; 11-1/2 mi. S and 2 mi. E Robertson, 9200 ft., 1; 14 mi. S and 2 mi. E Robertson, 9000 ft., 1.

Utah: *Uintah County:* Paradise Park, 21 mi. W, 15 mi. N Vernal, 10,050 ft., 1.

Additional, marginal records (Durrant, Univ. Kansas Publ., Mus. Nat. Hist., 6:356, August 19, 1952).--**Utah:** *Rich Co.:* Monte Cristo, 18 mi. W Woodruff, 8000 ft. *Salt Lake Co.:* Emigration Canyon, 8 mi. above forks, 6,000 ft.; Silver Lake Post Office (Brighton), 9,500 ft. *Wasatch Co.:* Wolf Creek Summit, 9,800 ft. *Daggett Co.:* Beaver Dams, 10,500 ft.

***Clethrionomys gapperi gauti*, new subspecies**

Type.--Male, adult, skin and skull; No. 133515, Biological Surveys Collections, United States National Museum, from Twining, 10,700 ft., Taos County, New Mexico; obtained on August 7, 1904, by James H. Gaut, original number 3086.

Range.--The Rocky Mountains of north-central New Mexico and south-central Colorado.

Diagnosis.--A brightly colored *Clethrionomys gapperi*; dorsal stripe near Chestnut (capitalized color terms after Ridgway, Color Standards and Color Nomenclature, Washington, D. C., 1912) with an underwash of near Light Ochraceous-Buff and a mixture of black-tipped guard hairs giving an over-all effect of between Tawny and Russet; braincase relatively large; zygomatic width and lambdoidal width large; nasals long.

Comparisons.--As compared with topotypes of *C. g. galei*, the color is lighter, the dorsal reddish stripe slightly narrower, sides brighter, with a wash of Light Ochraceous-Buff, grading ventrally into a slight wash of Pale Ochraceous-Buff, instead of a silvery-white venter characteristic of *C. g. galei*; zygomatic and lambdoidal breadths are greater, nasals slightly shorter, auditory bullae slightly more inflated, teeth larger, and braincase larger.

As compared with topotypes of *C. g. limitis*, *C. g. gauti* is darker, has a greater zygomatic breadth, longer upper tooth-row, longer nasals, and narrower rostrum.

Measurements.--External and cranial measurements of the type, and the average and extreme measurements of four adult males and one adult female from the type locality (including the type) and five miles south of the type locality are: Total length, 144, 147 (140-152); tail, 40, 42 (39-45); hind foot, 20, 19.3 (19-20); condylobasilar length, 22.3, 22.9 (22.2-24.0); zygomatic breadth, 13.6, 13.7 (13.5-14.0); lambdoidal breadth, 11.9, 11.7 (11.4-12.0); alveolar length of upper cheek-teeth, 5.1, 5.2 (5.1-5.4); interorbital breadth, 4.0, 3.9 (3.8-4.0); length of nasals, 7.0, 7.2 (7.0-7.6); breadth of rostrum, 2.9, 3.1 (2.9-3.4); length of incisive foramina, 4.8, 5.1 (4.8-5.3).

Remarks.--Two specimens from a locality 21 mi. W and 3 mi. N Saguache, Saguache County, Colorado, although referred to this subspecies on the basis of paler pelage, inflation of auditory bullae, and heavier teeth, show characters of *C. g. galei* in the narrowness across the zygomata and lambdoidal crest. Four specimens from Silverton (1 adult and 3 young adults) are referable to this subspecies on the basis of color of pelage and cranial proportions but are smaller than either *C. g. gauti* or *C. g. galei*.

The specimen from Pecos Baldy, Pecos Mountain, San Miguel County, New Mexico, referred by Bailey (N. Amer. Fauna, 52:192) to *Clethrionomys gapperi galei*, is here referred to *C. g. gauti* on geographical grounds.

The name *gauti* is proposed in honor of the collector of the type specimen, James H. Gaut.

Specimens examined.--Total, 14, distributed as follows and, unless otherwise stated, in the Biological Surveys Collection:

Colorado: *Saguache County:* 21 mi. W and 3 mi. N Saguache, N 38°, 106° 31', 9100 ft., 2 (Univ. Kansas Mus. Nat. Hist.). *San Juan Co.:* Silverton, 4.

New Mexico: *Taos County:* Twining, 10,700 ft., 3; 5 mi. S Twining, 11,400 ft., 3. *Sandoval County:* Goat Peak, Jemez Mountains, 1. *Colfax County:* 15 mi. SW Cimarron, 9000 ft., 1 (Amer. Mus. Nat. Hist.).

Additional records.--**New Mexico:** *San Miguel Co.:* Pecos Baldy, Pecos Mountain, 1 (Bailey, N. Amer. Fauna, 52:192, 1932).

***Clethrionomys gapperi limitis* (Bailey)**

1913. *Evotomys limitis* Bailey, Proc. Biol. Soc. Washington, 26:133, May 21.

Type locality.--Willow Creek, a branch of the Gilita, 8500 ft., Mogollon Mountains, Catron County, New Mexico.

Range.--Known from the Mogollon, San Mateo, and Magdalena mountains of western New Mexico.

Remarks.--Bailey (Proc. Biol. Soc. Washington, 26:133, May 21, 1913) described this animal as a species and gave as general characteristics: "Size slightly larger than *E. [= Clethrionomys gapperi] galei*; colors duller, grayer and less buffy; skull and dentition heavier." He further characterized the skull as: "Larger, heavier and conspicuously more ridged than in *galei*; bullae large and especially deep; dentition heavy throughout." The type of *C. limitis*, as judged from the measurements given by Bailey (*loc. cit.*), is an exceptionally old male.

Our comparison of six adult topotypes with a series of *C. g. galei* from Wyoming (18 adults from 3 mi. SSE Browns Peak, 10,000 ft., Albany County, in Univ. Kansas Mus. Nat. Hist.) and with three near-topotypes of *C. g. galei* (3 mi. S Ward, 9000 ft., Boulder County, Colorado, in Univ. Kansas Mus. Nat. Hist.) revealed that most of the differences noted by Bailey (*loc. cit.*) are not evident when individuals of comparable ages are examined. Some specimens of *C. g. galei* exceed *C. limitis* in ridging of the skull and size of the teeth although conspicuous ridges and large teeth are supposedly distinctive of *C. limitis*. The bullae, although averaging larger in *C. limitis*, can be matched in size by those of specimens of *C. g. galei* from Wyoming.

The differences evident between *C. limitis* and *C. g. galei* are of the kind and degree that serve to separate subspecies in the species *Clethrionomys gapperi* and, although actual evidence of intergradation is lacking, we think

that the relationships of *limitis* are better expressed by arranging it as a subspecies of *C. gapperi* than by retaining it as a full species.

Specimens examined.--Total, 7, all in the Biological Surveys Collection, distributed as follows:

New Mexico: *Catron County:* Willow Creek, 8500 ft., Mogollon Mountains, 4. *Socorro County:* San Mateo Peak, 10,000 ft., San Mateo Mountains, 2; Copper Canyon, 9000 ft., Magdalena Mountains, 1.

***Clethrionomys gapperi arizonensis*, new subspecies**

Type.--Female, adult, skin and skull; No. 158401, Biological Surveys Collection, United States National Museum; from Little Colorado River, 8300 ft., White Mountains, Apache County, Arizona; obtained September 12, 1908, by C. Birdseye, original number 152.

Range.--Known only from the White Mountains of eastern Arizona.

Diagnosis.--Dorsal stripe near Chestnut, with an underwash of between Tawny and Russet, and a mixture of black-tipped hairs, resulting in an overall effect of near Chestnut. Skull wide across zygomatic arches and narrow across mastoids; rostrum narrow and posterior border of palate straight.

Comparisons.--This subspecies needs close comparison only with the adjacent subspecies *C. g. limitis*. As compared with topotypes of *limitis*, *C. g. arizonensis* has darker pelage, narrower rostrum, greater width across zygomatic arches, lesser lambdoidal breadth, longer nasals, wider palate, and more inflated auditory bullae. The posterior border of the hard palate is straight in five skulls of the series that are complete (two skulls have the palatal regions broken); all *C. g. limitis* examined have a median posterior projection on the posterior border of the hard palate.

Measurements.--External and cranial measurements of the type, and the average and extreme measurements of three adult males and two adult females from the type locality (including the type) are: Total length, 160, 145.6 (137-160); tail, 44, 40.8 (37-46); hind foot, 18.5, 19.3 (18-20); condylobasilar length, 23.3, 22.8 (22.1-23.5); zygomatic breadth, 13.8, 13.4 (12.6-13.8); lambdoidal breadth, 11.5, 11.4 (11.0-11.6); alveolar length upper cheek-teeth, 5.5, 5.4 (5.2-5.5); interorbital breadth, 3.8, 3.9 (3.8-4.0); length of nasals, 7.6, 7.1 (6.9-7.6); breadth of rostrum, 3.1, 3.1 (3.0-3.2); length of incisive foramina, 5.5, 5.2 (5.0-5.5).

Remarks.--Hall and Davis (Proc. Biol. Soc. Washington, 47:55, February 9, 1934) reported 12 specimens of red-backed mice from Hannagan Meadow, 9500 to 9600 ft., and ten from Hannagan Creek, 8600 ft., all in Greenlee County, Arizona. Although they pointed out most of the cranial differences here described as diagnostic of *C. g. arizonensis*, they did not name the animals as new since they had no seasonally comparable materials;

thus they were unable to evaluate the differences noted in pelage. We have not examined the material referred to by Hall and Davis (*loc. cit.*), but, on the basis of their description, here refer it to *C. g. arizonensis*.

Specimens examined.--Total, 7, all from the type locality and all in the Biological Surveys Collection in the United States National Museum.

Grateful acknowledgment is made of the opportunity to study the specimens from New Mexico and Arizona in the Biological Surveys Collection of the United States National Museum and the material from South Dakota in the University of Michigan Museum of Zoology, as well as for the financial support afforded one of us (Cockrum) by the University of Kansas from its Research appropriation. Cockrum's work was part of a larger investigation of the geographic distribution of all North American native mammals, aided by a contract between the Office of Naval Research, Department of the Navy, and the University of Kansas (NR. 161-791). Also, assistance with some of field work was given by the Kansas University Endowment Association.

Transmitted June 21, 1952.



24-4369

*** END OF THE PROJECT GUTENBERG EBOOK GEOGRAPHIC
VARIATION IN RED-BACKED MICE (GENUS CLETHRIONOMYS)
OF THE SOUTHERN ROCKY MOUNTAIN REGION ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project GutenbergTM License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

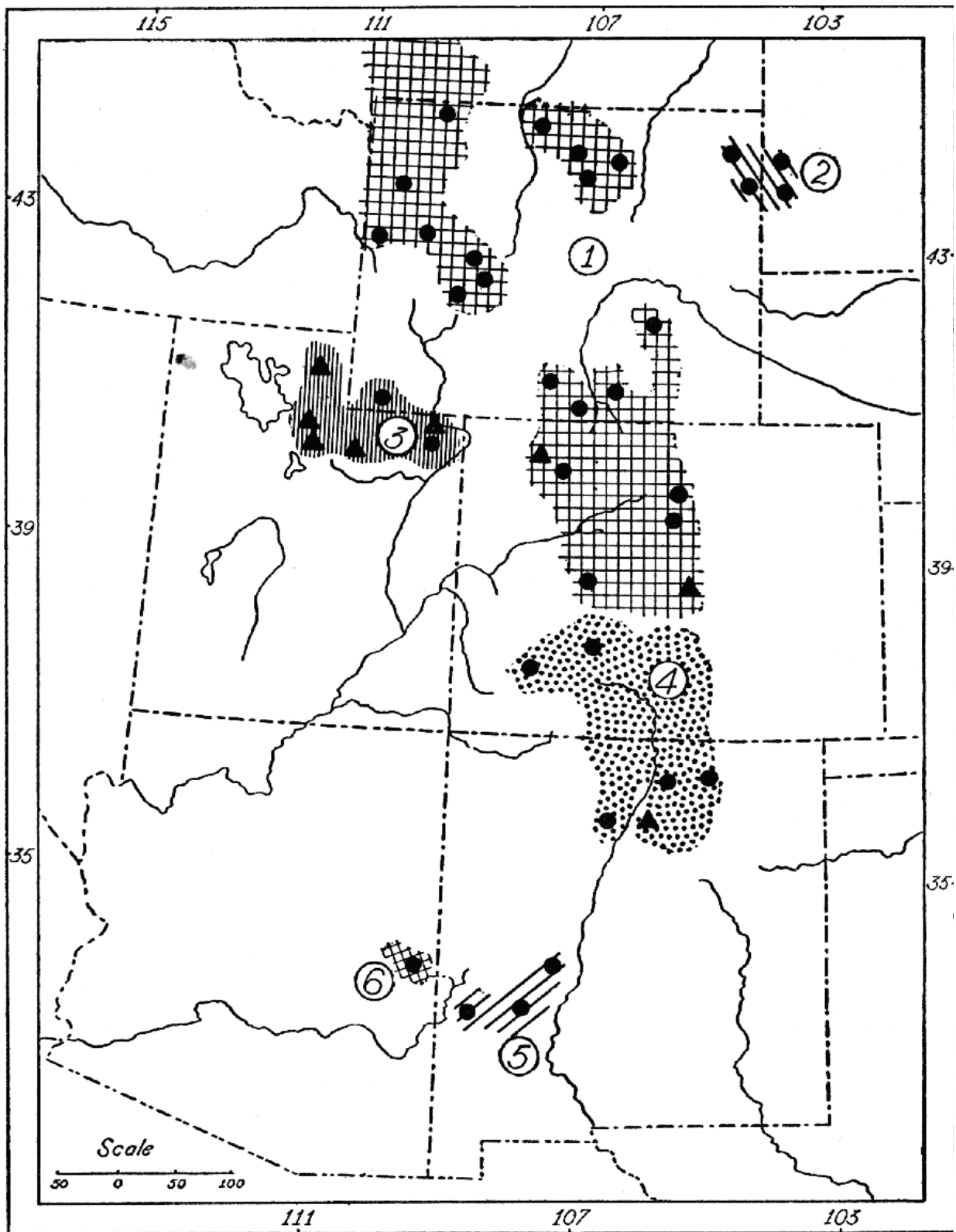
Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.



[back](#)